

TECNOLÓGICO NACIONAL DE MÉXICO INSTITUTO TECNOLÓGICO DE VERACRUZ

INGENIERÍA EN SISTEMAS COMPUTACIONALES

Práctica 2:

Creación de un Cluster en MongoDB Atlas usando Atlas CLI

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Asignatura:

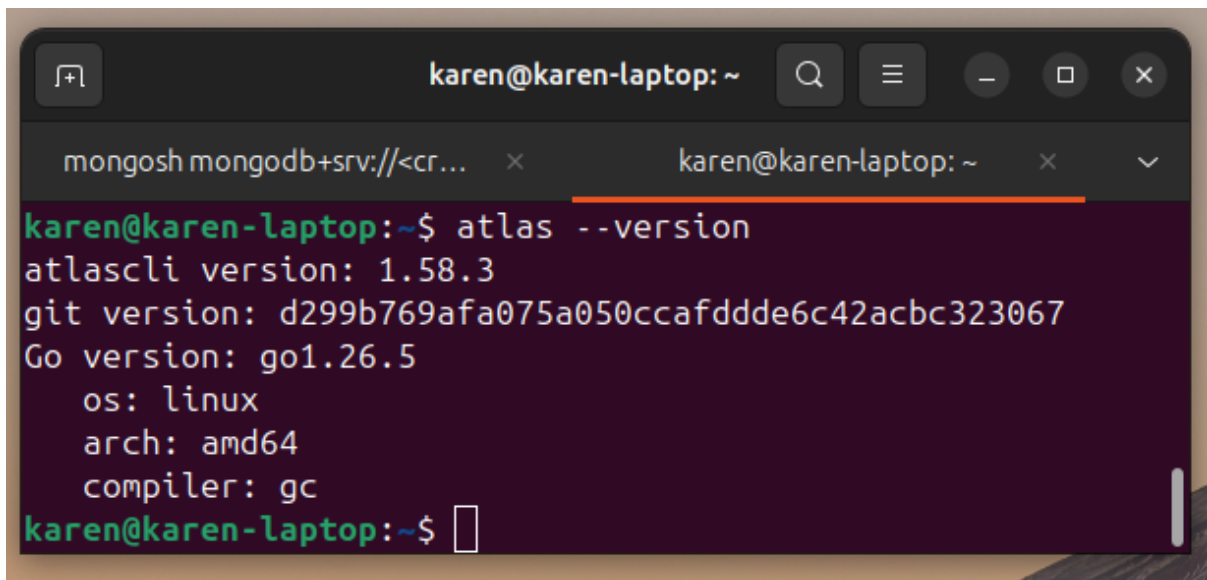
Base de Datos NoSQL



Resumen

En esta práctica se utilizó la Atlas CLI para crear y administrar un cluster de MongoDB Atlas desde la terminal. Primero se autenticó la CLI con `atlas auth login`. Al ejecutar `atlas setup`, Atlas rechazó la creación del cluster porque el proyecto original ya tenía un cluster gratuito (el que cree en clase pero olvidé tomar las capturas), por lo que se creó un proyecto nuevo (Practica2-CLI) y se repitió el setup sobre él. Se configuró el cluster MongoK en AWS (región US_EAST_1) con datos de muestra, un usuario de base de datos y la IP pública autorizada, y se verificó la conexión con `mongosh`. Después, con `mongodump` se exportó la base `zoologico` (colección `animales`, 5 documentos) desde el contenedor local de Podman, y con `mongorestore` se importó al cluster MongoK. Finalmente, se comprobó desde `mongosh` que los 5 documentos estaban en la nube.

Procedimiento

A terminal window titled 'karen@karen-laptop: ~' with a search icon, a menu icon, and window control buttons. It shows a command prompt where the user has entered 'atlas --version'. The output displays the versions of atlascli (1.58.3), git (d299b769afa075a050ccafddde6c42acbc323067), and Go (go1.26.5), along with the operating system (linux), architecture (amd64), and compiler (gc).

```
karen@karen-laptop:~$ atlas --version
atlascli version: 1.58.3
git version: d299b769afa075a050ccafddde6c42acbc323067
Go version: go1.26.5
  os: linux
  arch: amd64
  compiler: gc
karen@karen-laptop:~$
```

Creación y configuración inicial del Cluster

```
mongosh mongodb+srv://<credentials>@mongok.dksnllp.mongodb.net/
mongosh mongodb+srv://<credentials>@mo... x karen@karen-laptop: ~
karen@karen-laptop:~$ atlas auth login
? Select authentication type: UserAccount

To verify your account, copy your one-time verification code:

3VYC-W74X

Paste the code in the browser when prompted to activate your Atlas CLI. Your code will expire after 3 minutes.

To continue, go to https://account.mongodb.com/account/connect

Press Enter to open the browser and complete authentication...
[77051:77051:0927/164935.848009:ERROR:ui/ozone/platform/wayland/gpu/wayland_surface_factory.cc:249] '--ozone-platform=wayland' is not compatible with Vulkan. Consider switching to '--ozone-platform=x11' or disabling Vulkan
Created TensorFlow Lite XNNPACK delegate for CPU.
Successfully logged in as l22020987@veracruz.tecnm.mx.
? Default Output Format: json
```

```
mongosh mongodb+srv://<credentials>@mongok.dksnllp.mongodb.net/
mongosh mongodb+srv://<credentials>@mo... x karen@karen-laptop: ~
karen@karen-laptop:~$ atlas setup

[Default Settings]
Cluster Name: Cluster49512
Cloud Provider and Region: AWS - US_EAST_1
Database User Username: Cluster49512
Load sample data: Yes
Allow connections from (IP Address): 187.241.14.39

? How do you want to set up your Atlas cluster? custom

Press [Enter] to use the default values.

Enter [?] on any option to get help.

[Set up your Atlas cluster]

? Cluster Name [This can't be changed later] Mongok
```

```
mongosh mongodb+srv://<credentials>@mongok.dksnllp.mongodb.net/
mongosh mongodb+srv://<credentials>@mo... x karen@karen-laptop: ~
[Set up your Atlas cluster]

? Cluster Name [This can't be changed later] MongoK

? Cloud Provider AWS

? Cloud Provider Region US_EAST_1

? Do you want to load a sample dataset? Yes

[Set up your database authentication access details. Store them in a secure location.]

? Database User Username karen_cli

? Database User Password [You must include >10 characters. Press Enter to use an auto-generated password] XmxQT1BkVHZE

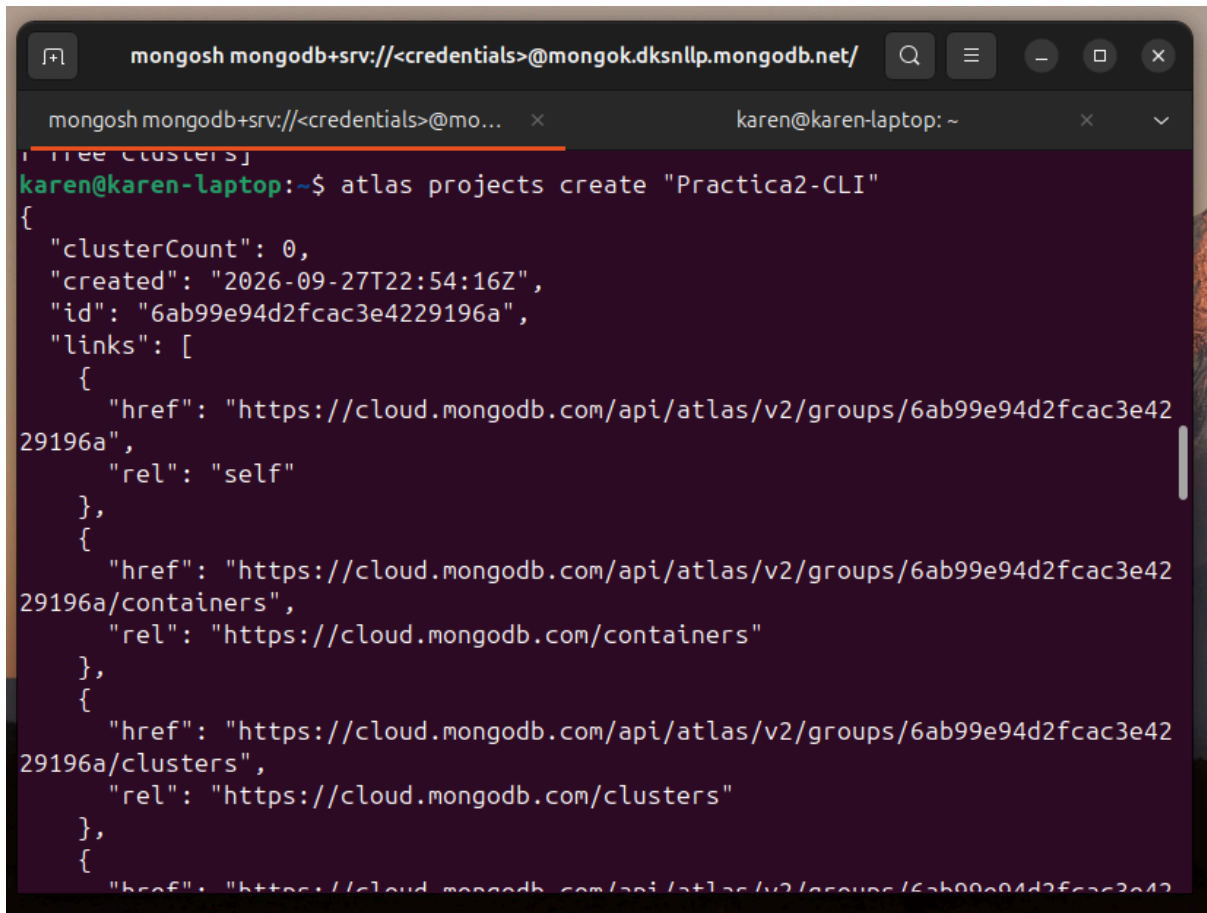
[Set up your database network access details]

? IP Access List Entry [Press Enter to use your public IP address '187.241.14.3
```

Error mencionado por tener un cluster ya existente

```
mongosh mongodb+srv://<credentials>@mongok.dksnllp.mongodb.net/
mongosh mongodb+srv://<credentials>@mo... x karen@karen-laptop: ~
? IP Access List Entry [Press Enter to use your public IP address '187.241.14.39'] 187.241.14.39
Detected clusterWideScaling mode. If you require more flexibility in shard scaling, consider using --autoScalingMode independentShardScaling.
Error: https://cloud.mongodb.com/api/atlas/v2/groups/6ab598153e686486c0f028b5/clusters POST: HTTP 400 Bad Request (Error code: "CANNOT_CREATE_FREE_CLUSTER_VIA_PUBLIC_API") Detail: Cannot create a cluster with instance size M0 via the public api: This project has reached the limit for the number of free clusters. Reason: Bad Request. Params: [This project has reached the limit for the number of free clusters]
karen@karen-laptop:~$ atlas projects create "Practica2-CLI"
{
  "clusterCount": 0,
  "created": "2026-09-27T22:54:16Z",
  "id": "6ab99e94d2fcac3e4229196a",
  "links": [
    {
      "href": "https://cloud.mongodb.com/api/atlas/v2/groups/6ab99e94d2fcac3e4229196a",
      "rel": "self"
    },
    {
      "href": "https://cloud.mongodb.com/api/atlas/v2/groups/6ab99e94d2fcac3e42
```

crear otro proyecto

A terminal window with a dark purple background. The prompt is 'karen@karen-laptop:~\$'. The command 'atlas projects create "Practica2-CLI"' has been entered. The output is a JSON object representing the created project. The terminal window has a title bar with 'mongosh mongodb+srv://<credentials>@mongok.dksnllp.mongodb.net/' and standard window controls. There are also browser tabs visible at the top.

```
mongosh mongodb+srv://<credentials>@mongok.dksnllp.mongodb.net/
karen@karen-laptop:~$ atlas projects create "Practica2-CLI"
{
  "clusterCount": 0,
  "created": "2026-09-27T22:54:16Z",
  "id": "6ab99e94d2fcac3e4229196a",
  "links": [
    {
      "href": "https://cloud.mongodb.com/api/atlas/v2/groups/6ab99e94d2fcac3e4229196a",
      "rel": "self"
    },
    {
      "href": "https://cloud.mongodb.com/api/atlas/v2/groups/6ab99e94d2fcac3e4229196a/containers",
      "rel": "https://cloud.mongodb.com/containers"
    },
    {
      "href": "https://cloud.mongodb.com/api/atlas/v2/groups/6ab99e94d2fcac3e4229196a/clusters",
      "rel": "https://cloud.mongodb.com/clusters"
    },
    {
      "href": "https://cloud.mongodb.com/api/atlas/v2/groups/6ab99e94d2fcac3e4229196a/clusters"
    }
  ]
}
```

```
mongosh mongodb+srv://<credentials>@mongok.dksnllp.mongodb.net/
mongosh mongodb+srv://<credentials>@mo... x karen@karen-laptop: ~
karen@karen-laptop:~$ atlas setup --projectId 6ab99e94d2fcac3e4229196a

[Default Settings]
Cluster Name: Cluster49699
Cloud Provider and Region: AWS - US_EAST_1
Database User Username: Cluster49699
Load sample data: Yes
Allow connections from (IP Address): 187.241.14.39

? How do you want to set up your Atlas cluster? custom

Press [Enter] to use the default values.

Enter [?] on any option to get help.

[Set up your Atlas cluster]
? Cluster Name [This can't be changed later] MongoK
? Cloud Provider AWS
```

```
mongosh mongodb+srv://<credentials>@mongok.dksnllp.mongodb.net/
mongosh mongodb+srv://<credentials>@mo... x karen@karen-laptop: ~
? Cloud Provider Region US_EAST_1
? Do you want to load a sample dataset? Yes

[Set up your database authentication access details. Store them in a secure location.]
? Database User Username karen_cli
? Database User Password [You must include >10 characters. Press Enter to use an auto-generated password] aXheTXhiV0x6

[Set up your database network access details]
? IP Access List Entry [Press Enter to use your public IP address '187.241.14.39'] 187.241.14.39
Detected clusterWideScaling mode. If you require more flexibility in shard scaling, consider using --autoScalingMode independentShardScaling.
We are deploying MongoK...

Store your database authentication access details in a secure location:
```

```
mongosh mongodb+srv://<credentials>@mongok.dksnllp.mongodb.net/
Store your database authentication access details in a secure location:
Database User Username: karen_cli
Database User Password: aXheTXhiV0x6

Creating your cluster... [It's safe to 'Ctrl + C']
Cluster created.
Your connection string: mongodb+srv://mongok.dksnllp.mongodb.net

Loading sample data into your cluster... [It's safe to 'Ctrl + C']
? How would you like to connect to your cluster? mongosh

Current Mongosh Log ID: 6ab99fce0eff2d41133c7297

Connecting to:      mongodb+srv://<credentials>@mongok.dksnllp.mongodb.net/
?appName=mongosh+2.10.0
Using MongoDB:      8.0.32
Using Mongosh:      2.10.0

mongosh 2.12.0 is available for download: https://www.mongodb.com/try/download/shell
```

Creado con éxito :D

```
mongosh mongodb+srv://<credentials>@mongok.dksnllp.mongodb....
mongosh 2.12.0 is available for download: https://www.mongodb.com/try/downlo
ad/shell

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

Atlas atlas-dc8tzt-shard-0 [primary] test> show dbs
sample_airbnb      52.79 MiB
sample_analytics   9.56 MiB
sample_geospatial 1.31 MiB
sample_guides      40.00 KiB
sample_mflix       123.42 MiB
sample_restaurants 6.44 MiB
sample_supplies    1.05 MiB
sample_training    47.50 MiB
sample_weatherdata 2.60 MiB
admin              0 B
local              0 B
Atlas atlas-dc8tzt-shard-0 [primary] test> 
```

Importar una base de datos ya existente, en este caso se agregó la de la práctica 1.

```
karen@karen-laptop: ~  
mongosh mongodb+srv://<credentials>@... x karen@karen-laptop: ~  
karen@karen-laptop:~$ mongorestore --uri="mongodb+srv://mongok.dksnllp.mongodb.net/" --username karen_cli --nsInclude="zoologico.*" ~/backup_zoologico  
Enter password for mongo user:  
  
2026-09-27T18:40:01.487-0600 preparing collections to restore from  
2026-09-27T18:40:01.488-0600 reading metadata for zoologico.animales from  
/home/karen/backup_zoologico/zoologico/animales.metadata.json  
2026-09-27T18:40:01.785-0600 restoring zoologico.animales from /home/karen/backup_zoologico/zoologico/animales.bson  
2026-09-27T18:40:01.873-0600 finished restoring zoologico.animales (5 documents, 0 failures)  
2026-09-27T18:40:01.873-0600 no indexes to restore for collection zoologico.animales  
2026-09-27T18:40:01.873-0600 5 document(s) restored successfully. 0 document(s) failed to restore.
```

```
karen@karen-laptop: ~  
mongosh mongodb+srv://<credentials>@... x karen@karen-laptop: ~ x v  
karen@karen-laptop:~$ mongosh "mongodb+srv://mongok.dksnllp.mongodb.net/" --  
username karen_cli --eval "db.getSiblingDB('zoologico').animales.find().toAr  
ray()"  
Enter password: *****  
[  
  {  
    _id: ObjectId('6ab8a77962b50565b982a612'),  
    nombre: 'León',  
    especie: 'Panthera leo',  
    habitat: 'Sabana',  
    dieta: 'Carnívoro',  
    peso_kg: 190  
  },  
  {  
    _id: ObjectId('6ab8a77962b50565b982a613'),  
    nombre: 'Elefante africano',  
    especie: 'Loxodonta africana',  
    habitat: 'Sabana',  
    dieta: 'Herbívoros',  
    peso_kg: 5400  
  },  
  {  
    _id: ObjectId('6ab8a77962b50565b982a614'),  
    nombre: 'Pingüino emperador',  
    especie: 'Aptenodytes forsteri',  
    habitat: 'Antártida',  
    dieta: 'Carnívoro',  
    peso_kg: 30  
  },  
  {  
    _id: ObjectId('6ab8a77962b50565b982a615'),  
    nombre: 'Jaguar',  
    especie: 'Panthera onca',  
    habitat: 'Selva',  
    dieta: 'Carnívoro',  
    peso_kg: 90  
  },  
  {  
    _id: ObjectId('6ab8a77962b50565b982a616'),  
    nombre: 'Panda gigante',  
    especie: 'Ailuropoda melanoleuca',  
    habitat: 'Bosque de bambú',  
    dieta: 'Herbívoros',  
    peso_kg: 100  
  }  
]
```

Comandos usados

Comando	Descripción
<code>atlas -version</code>	Muestra la versión instalada de la Atlas CLI
<code>atlas auth login</code>	Autentica la CLI con tu cuenta de Atlas mediante un código de verificación en el navegador
<code>atlas setup</code>	Asistente interactivo que crea cluster, usuario de base de datos, acceso por IP y datos de muestra
<code>atlas projects create "Practica2-CLI"</code>	Crea un proyecto nuevo en la organización (necesario por el límite de un cluster gratuito por proyecto)
<code>atlas setup --projectId 6ab99e94d2fcac3e4229196a</code>	Ejecuta el asistente de creación de cluster dentro del proyecto nuevo
<code>show dbs</code>	Lista las bases de datos del cluster (dentro de mongosh)
<code>mongodump --version / mongorestore --version</code>	Verifican la instalación de las Database Tools
<code>mongodump --host 127.0.0.1 --port 27018 --db zoologico --out ~/backup_zoologico</code>	Exporta la base zoologico del contenedor local a archivos BSON
<code>ls -R ~/backup_zoologico</code>	Lista los archivos generados por el respaldo
<code>mongorestore --uri="mongodb+srv://mongok.dksnllp.mongodb.net/" --username karen_cli --nsInclude="zoologico.*" ~/backup_zoologico</code>	Importa el respaldo al cluster MongoK en Atlas (pide la contraseña de forma interactiva)